

Wrocław University of Science and Technology
Faculty of Management

Business Intelligence Workplace

Second task set report

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1 Descriptive analysis

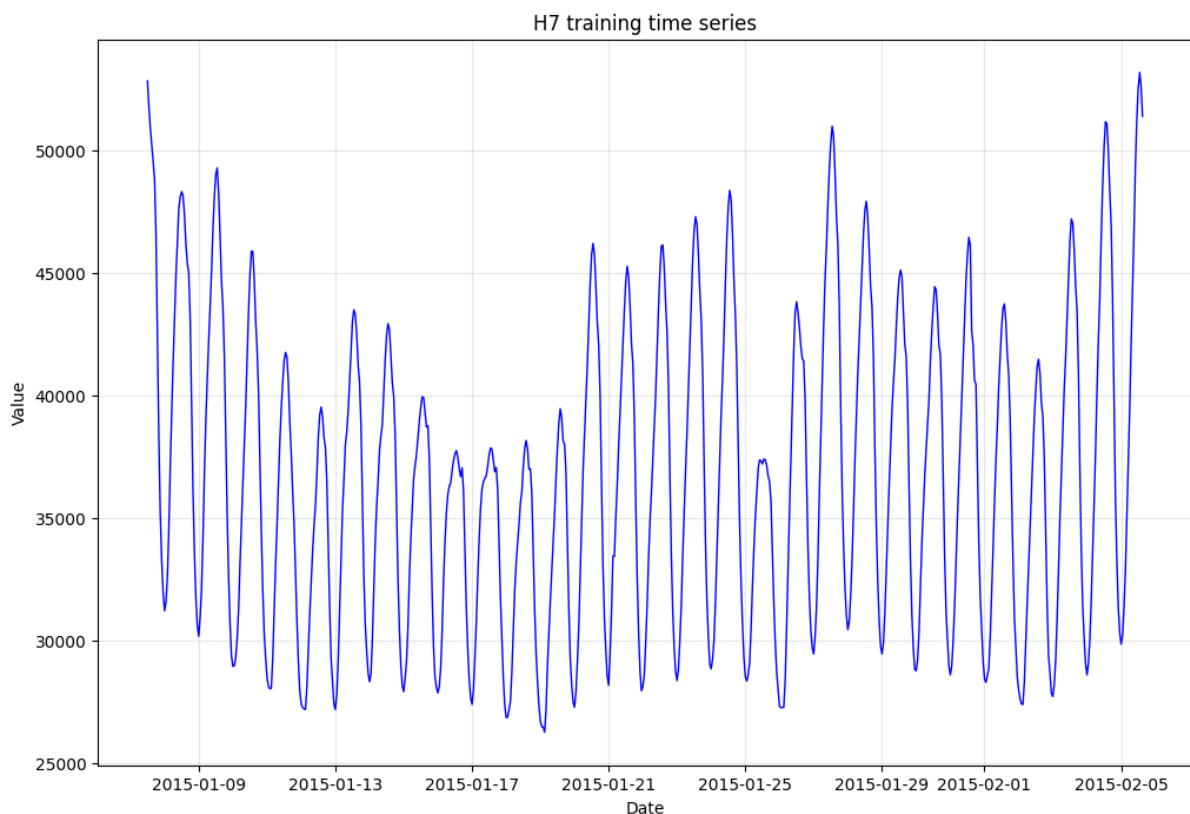
1.1 Task description

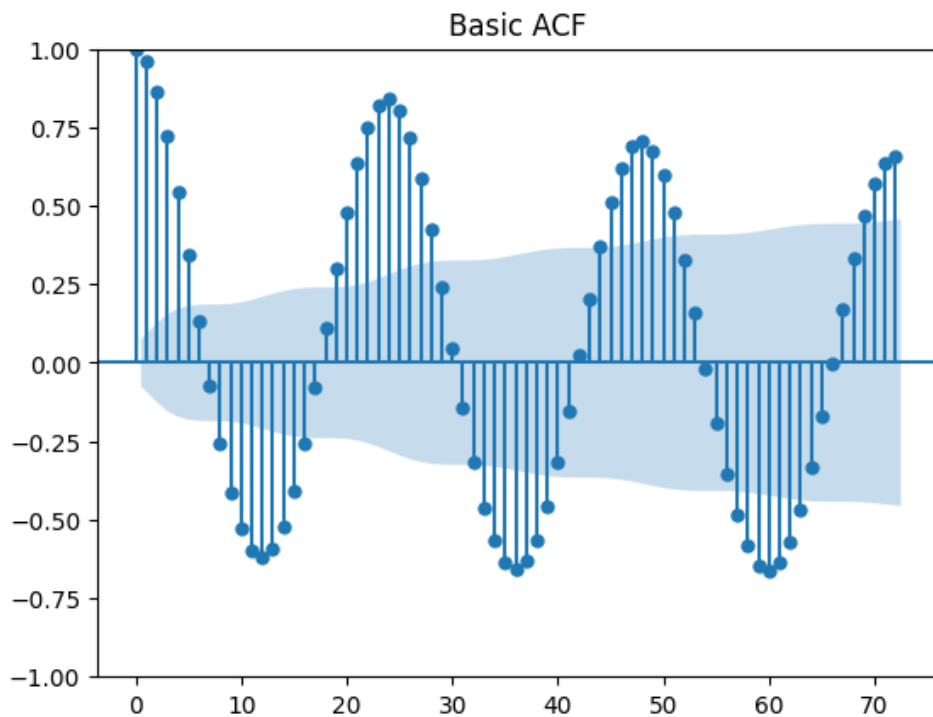
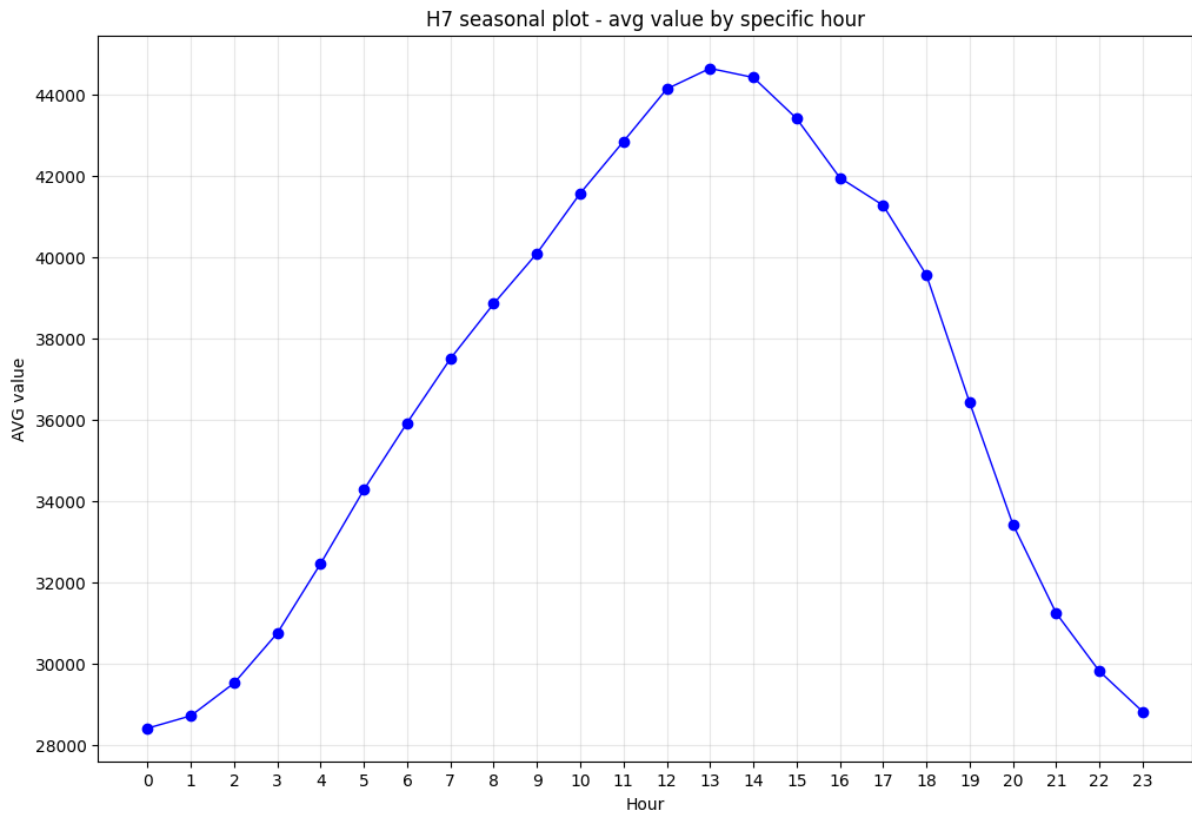
The objective of this task was to provide an initial descriptive overview of the assigned H7 hourly time series from the M4 Forecasting Competition Dataset from Kaggle. This involved visualizing the raw training data to identify fundamental structural characteristics, such as overarching long term trends, recurring daily seasonal patterns and any potential outliers, setting the foundational context required before applying formal mathematical modeling techniques.

1.2 Overview

count	700.000000
mean	36718.152857
std	6351.335330
min	26245.000000
25%	31075.750000
50%	36597.500000
75%	41536.000000
max	53168.000000

The descriptive analysis of the H7 training time series, which consists of 700 hourly observations, reveals a mean value of approximately 36,718 and a standard deviation of 6,351, fluctuating between a minimum of 26,245 and a peak of 53,168.





The initial "H7 training time series" plot provides insight into its overall historical trajectory and continuous variations over time, where clear seasonality occurs. To isolate daily cyclical behaviors, the seasonal plot aggregates the data to show the average value for each specific hour of the day, clearly illustrating and thus confirming the previous assumption of a strong daily seasonality. Finally, the ACF plot displays the correlation between the time series and its own lagged values to help identify patterns like seasonality or non-stationarity. Said plot characteristically reveals prominent, statistically significant spikes at precise 24 hour intervals. This

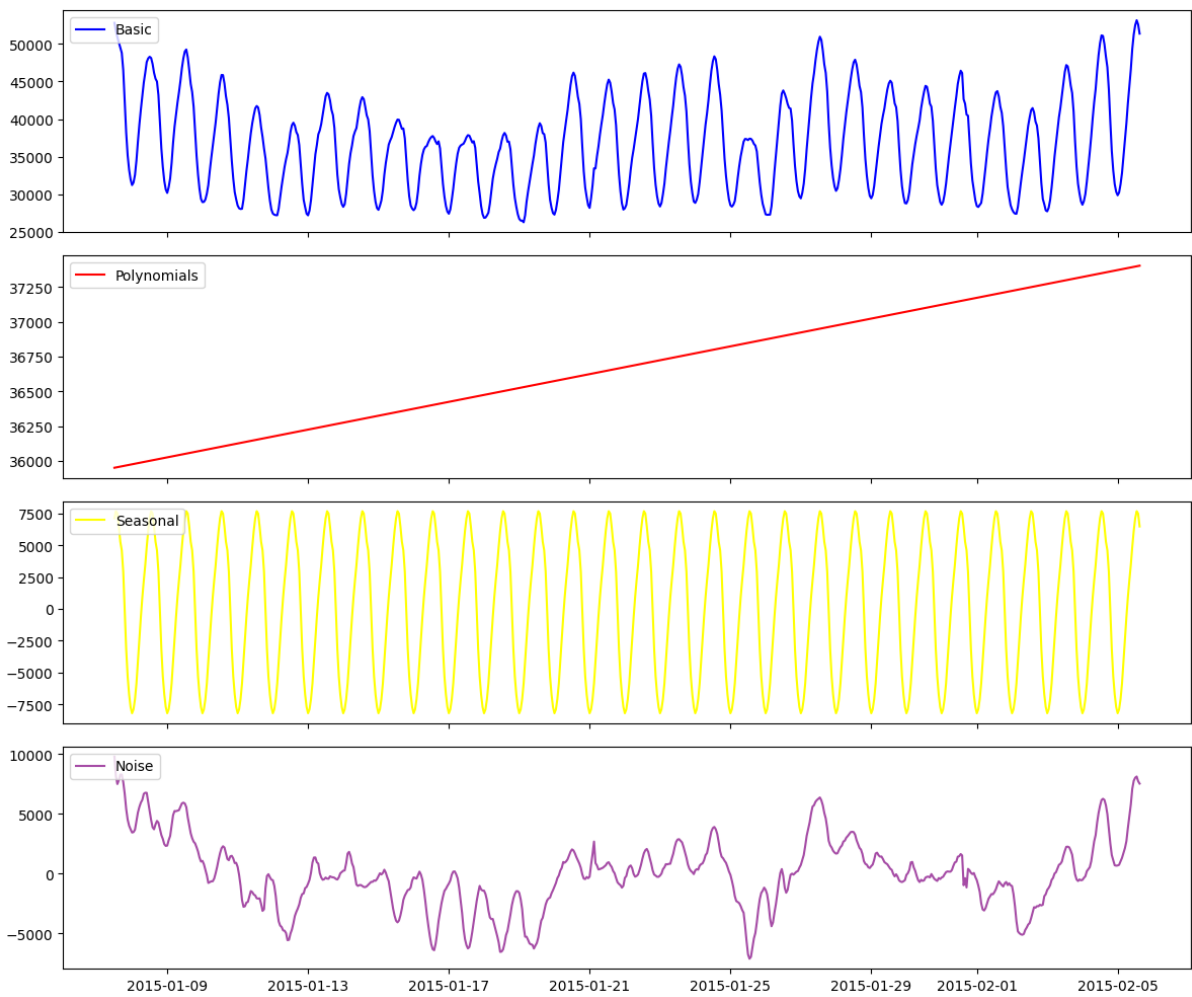
mathematically confirms the rigid day seasonality observed in the previous plots. Furthermore, the overall slow decay of the autocorrelation values, rather than a rapid drop to zero, is a classic hallmark of non-stationarity. It indicates that the series is heavily influenced by its historical values and an ongoing underlying trend, justifying the subsequent need for polynomial detrending and deseasonalization to isolate the stationary noise.

2 Classical decomposition

2.1 Task description

Classical decomposition breaks down the time series into its constituent trend, seasonal, and random noise components utilizing the additive model $X_t = m_t + s_t + Y_t$. The methodology required calculating an initial trend estimate via a 24 period centered moving average, extracting and normalizing the 24 hour seasonal component, and then applying a polynomial fit to the deseasonalized data to again estimate the final trend.

2.2 Overview



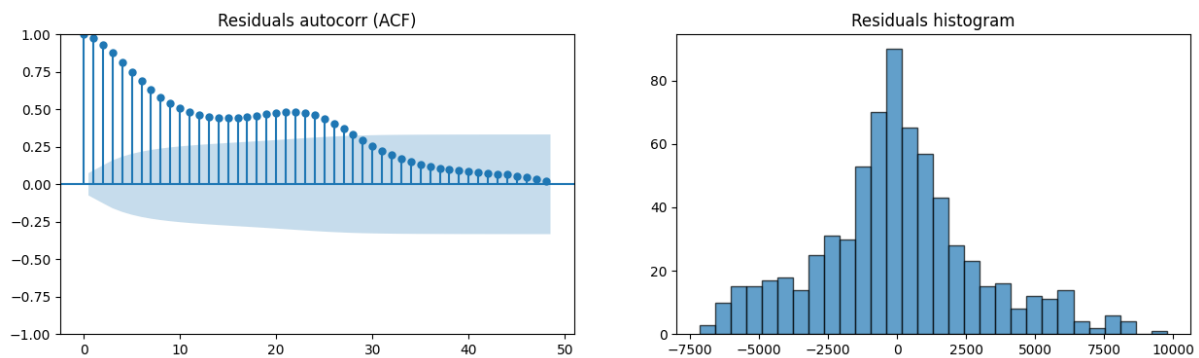
By extracting both a global, linear trend and a rigid daily seasonal cycle from the raw data, the analysis distills the complex series down to its underlying, stationary residuals. This linear polynomial fit confirms that the broad, steady historical movements and predictable daily rhythms account for the vast majority of the data's variance.

3 Residual analysis

3.1 Task description

Following the decomposition, this task evaluated the isolated noise series to determine whether the residuals behave as independent and identically distributed random variables. By employing visual autocorrelation plots and formal statistical testing, this analysis verified whether the classical decomposition successfully captured all deterministic patterns or if underlying temporal dependencies remain in the residuals that require further autoregressive modeling.

3.2 Overview



ACF and the residuals histogram plots critically evaluate the quality of the isolated noise component after detrending and deseasonalization. While the histogram demonstrates that the residuals are centered around zero, they do not form a normal distribution. The subsequent statistical summary reveals a high kurtosis (14.87) and a Jarque-Bera p-value of 0.00, confirming that the noise component is heavily tail heavy and contains irregular extreme shocks that deviate from normality.

--- Ljung-Box Test Results ---		
	lb_stat	lb_pvalue
24	6098.814252	0.0

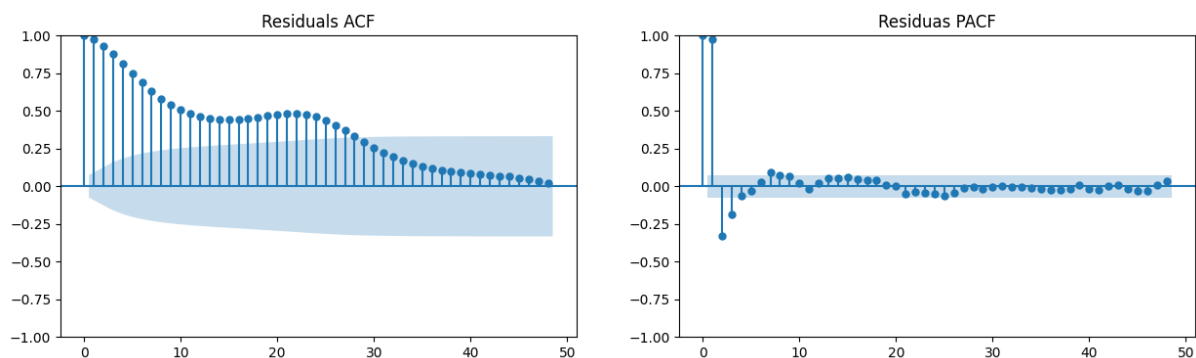
Previous visual evidence of remaining temporal dependence is statistically confirmed by the Ljung-Box test at lag 24, which yields a test statistic of 6098.81 value and a p-value of 0.0, definitively rejecting the H_0 hypothesis of nonexistent autocorrelation.

4 Time series model selection

4.1 Task description

Building upon the residual analysis, this task involved fitting an ARIMA model specifically to the isolated noise sequence. This step relied on inspecting autocorrelation and partial autocorrelation functions, alongside quantitative criteria such as the AIC, to systematically select the optimal combination of parameters capable of capturing any structural patterns left behind by the initial classical decomposition.

4.2 Overview



The dual ACF and PACF plots provide further insights into the internal structure of the residuals. Instead of remaining within the bounds of statistical insignificance, both plots display prominent initial spikes and complex trailing patterns that highlight lingering short-term temporal dependencies. This visual evidence signals the presence of overlapping autoregressive and moving average dynamics, explicitly dictating the necessity of a mixed-order modeling approach and directly justifying the subsequent grid search to calibrate the parameters needed to extract the remaining predictive signal.

SARIMAX Results						
=====						
Dep. Variable:	y	No. Observations:	700			
Model:	ARIMA(3, 0, 2)	Log Likelihood	-5061.032			
Date:	Tue, 26 May 2026	AIC	10136.064			
Time:	21:10:51	BIC	10167.921			
Sample:	01-07-2015	HQIC	10148.378			
	- 02-05-2015					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	0.0687	728.158	9.44e-05	1.000	-1427.094	1427.232
ar.L1	1.2207	0.119	10.251	0.000	0.987	1.454
ar.L2	-0.0041	0.217	-0.019	0.985	-0.429	0.420
ar.L3	-0.2446	0.105	-2.336	0.019	-0.450	-0.039
ma.L1	0.4771	0.119	4.009	0.000	0.244	0.710
ma.L2	0.2487	0.038	6.628	0.000	0.175	0.322
sigma2	1.107e+05	2805.345	39.452	0.000	1.05e+05	1.16e+05
=====						
Ljung-Box (L1) (Q):		1.03	Jarque-Bera (JB):	4122.48		
Prob(Q):		0.31	Prob(JB):	0.00		
Heteroskedasticity (H):		1.25	Skew:	-0.32		
Prob(H) (two-sided):		0.08	Kurtosis:	14.87		
=====						

The SARIMAX results reveal that an ARIMA(3, 0, 2) architecture optimally captures these residual dynamics, achieving the lowest AIC score (10,136.06). However, an analysis of the individual parameters indicates mixed statistical significance. While the AR(1) and MA(2) coefficients are highly significant ($p = 0.000$), the AR(2), AR(3),

and MA(1) terms fail to achieve significance at the standard 0.05 level. This suggests that while the overall model structure minimizes information loss according to the AIC, a simpler model might be more miserly.

5 Predictive analysis

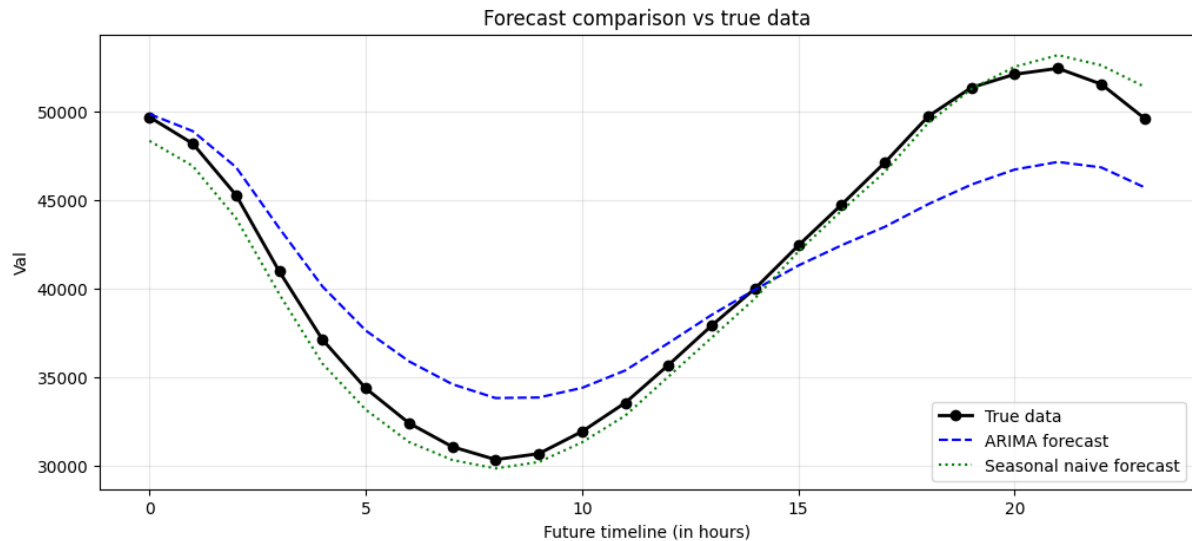
5.1 Task description

The final task synthesized the developed models to forecast the first 24 observations of the unseen test set. This was achieved by generating future predictions from the fitted ARIMA model and recombining them with extrapolated values from both the polynomial trend and the 24 hour seasonal component. The predictive performance of this combined approach is then rigorously evaluated through error metrics and compared against standard baseline techniques, namely the mean, naive, and seasonal naive forecasts.

5.2 Overview

Method	MAE	RMSE
ARIMA final model forecast	2822.2526	3259.2885
Mean forecast	7763.8240	9187.3778
Naive forecast	9869.8750	12418.1837
Seasonal naive forecast	810.5833	913.5967

The quantitative evaluation of the models over a 24 hour forecasting horizon starkly illustrates the role of cyclical patterns in the analyzed time series. By comparing MAE and RMSE, the results show that while the fully recombined composite ARIMA model easily outperforms flat baselines like the Mean and standard Naive forecasts, it falls drastically short of the Seasonal Naive approach. Achieving a remarkably low MAE, the Seasonal Naive model additionally proves the previously established conclusion of the time series being heavily dominated by its rigid hourly rhythm.



Graphically, the Seasonal Naive forecast almost entirely overlaps with the true data's trajectory, tightly mirroring every specific day peak and trough with high precision. Conversely, while the ARIMA forecast successfully captures the shape very similar to a wave and underlying trend of the series, it displays a noticeable vertical offset and smoothed deviations, failing to hit the exact localized extremes of the real cycle. This visual evidence beautifully reinforces the numerical metrics, confirming that for this specific temporal structure, cyclical repetition is practically superior to advanced statistical modeling.

6 Final remarks

The most profound takeaway from the dataset is how heavily the metric's behavior is dictated by a strict, predictable one day schedule (24 hours). The values shockingly do not fluctuate - rather, they follow a rigid daily rhythm that accounts for the vast majority of all movement within the data, which confirms that whatever phenomenon is being described by this set is heavily bound to the time of the day. While the daily rhythm defines the broad strokes of the data, the underlying deviations are highly revealing. When the predictable patterns are stripped away, the remaining fluctuations are not entirely random. The system is occasionally subjected to extreme, irregular spikes (up or down), and when they do occur, they do not vanish but leave a long lasting effect destabilizing system's future. If I had to point out the most valuable insight, that was obtained from exploring this dataset, it would be that data with such strong seasonality (in this case daily cycle) is not worth smoothing out, as it actually backfires. In such case, the best possible way to predict what will happen next is to simply mirror the specific hourly patterns of the days before which will yield the best results possible.